

Problem

Given that $F(\omega) = \mathcal{F}[f(t)]$, prove the following results, using the definition of Fourier transform:

(a) $\mathcal{F}[f(t - t_0)] = e^{-j\omega t_0} F(\omega)$

(b) $\mathcal{F}\left[\frac{df(t)}{dt}\right] = j\omega F(\omega)$

(c) $\mathcal{F}[f(-t)] = F(-\omega)$

(d) $\mathcal{F}[tf(t)] = j \frac{d}{d\omega} F(\omega)$

Step-by-step solution

Step 1 of 4

(a) $F[f(t - t_0)] = \int_{-\infty}^{\infty} f(t - t_0) e^{-j\omega t} dt$

Let $t - t_0 = \lambda \quad \longrightarrow \quad t = \lambda + t_0, \quad dt = d\lambda$

$$F[f(t - t_0)] = \int_{-\infty}^{\infty} f(\lambda) e^{-j\omega \lambda} e^{-j\omega t_0} d\lambda = e^{-j\omega t_0} F(\omega)$$

Step 2 of 4

(b) Given that $f(t) = F^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$

$$f'(t) = \frac{j\omega}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = j\omega F^{-1}[F(\omega)]$$

or

$$F[f'(t)] = j\omega F(\omega)$$

Step 3 of 4

(c) This is a special case of the time scaling property when $a = -1$. Hence,

$$F[f(-t)] = \frac{1}{|-1|} F(-\omega) = F(-\omega)$$

Step 4 of 4

$$(d) \quad F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

Differentiating both sides respect to ω and multiplying by t yields

$$j \frac{dF(\omega)}{d\omega} = j \int_{-\infty}^{\infty} (-jt) f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} tf(t) e^{-j\omega t} dt$$

Hence,

$$j \frac{dF(\omega)}{d\omega} = F[tf(t)]$$